



10 CFR 50.73

LG-19-094

September 20, 2019

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 2
Renewed Facility Operating License No. NPF-85
NRC Docket No. 50-353

Subject: LER 2019-002-00, Emergency Diesel Generator Lube Oil Pressure Sensing Line
Leak Resulting in a Condition Prohibited by Technical Specifications

Enclosed is a Licensee Event Report (LER) that addresses an Emergency Diesel Generator (EDG) being inoperable for greater than 30 days, resulting in a condition prohibited by Technical Specifications (TS) at Limerick Generating Station, Unit 2.

This LER is being submitted pursuant to the requirements of 10 CFR 50.73(a)(2)(i)(B).

There are no commitments contained in this letter.

If you have any questions, please contact Robert B. Dickinson at (610) 718-3400.

Respectfully,

Mark Boh for Frank Sturniolo

Frank Sturniolo
Vice President – Limerick Generating Station
Exelon Generation Company, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

1. Facility Name Limerick Generating Station Unit 2	2. Docket Number 05000 353	3. Page 1 OF 4
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4. Title
Emergency Diesel Generator Lube Oil Pressure Sensing Line Leak Resulting in a Condition Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
07	22	19	2019	002	00	09	20	2019	Facility Name	Docket Number 05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(ii)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(iii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact Robert Dickinson, Manager Regulatory Assurance	Telephone Number (Include Area Code) 610-718-3400
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
B	EK	PSF	NA	Yes					

14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date**

Month	Day	Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On 7/22/19, the instrumentation line threaded pipe connection for a pressure indicator, was found sheared during Emergency Diesel Generator (EDG) D24 slow start monthly run. The one-quarter inch instrumentation line threaded connection tees off the four-inch main lube oil piping near the discharge of the engine driven lube oil pump. Prior to the run there was no indication of a leak at this connection. This threaded pipe was pro-actively replaced on 5/23/18, and on 1/21/19 the threaded pipe connection failed due to high-cyclic fatigue. The corrective actions to address the failure on 1/21/19 were to install a support on the threaded connection and repair a broken U-bolt. These actions did not resolve the underlying condition. The cause of the failure on 7/22/19 was high-cyclic fatigue experienced at the threaded pipe connection. The corrective action is to redesign the connection to reduce the stress on the threaded pipe connection. In the interim, the instrument line was removed from service.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Limerick Generation Station, Unit 2	05000- 353	19	002	00

NARRATIVE**Unit Condition Prior to the Event**

Limerick Generating Station (LGS), Unit 2 was operating in Operational Condition (OPCON) 1 at 100 percent power at the time of the event. There were no other structures, systems, or components inoperable that contributed to the event.

Description of the Event

On 7/22/19, an instrumentation line threaded pipe connection for a pressure indicator, was found sheared during the Emergency Diesel Generator (EDG) D24 [EIS:EK] slow start monthly run. The one-quarter inch instrumentation line threaded connection tees off the four-inch main lube oil piping near the discharge of the engine driven lube oil pump.

EDG D24 was started at 21:01, loaded at 21:22, and fully loaded at 21:33. A dirty drip pan alarm came in at 22:05 and subsequently the lube oil leak was identified. The discovery time of the leak at approximately 22:05 is very close to the actual failure time of the threaded pipe. Prior to the failure, there were no leaks or degraded conditions present.

There have been leaks identified at this connection. On 3/21/2013, a 1-2 drop per minute (DPM) leak was identified on the same threaded connection. One month later, on 4/27/2013, the EDG was made inoperable when a threaded connection failed and the 1-2 DPM leak became a 140 DPM leak. The investigation performed for this failure determined the cause to be high-cycle fatigue from engine vibration exacerbated by a missing grommet on one of the downstream supports. The missing grommet was re-installed, and the threaded pipe was replaced. For approximately six years following this repair, no additional leakage occurred at this location.

On 12/12/16, a leak developed on this connection on EDG D13. The investigation determined that the cause was cyclic fatigue of the threaded connection with a contributing factor of manipulation of the valve downstream of the threaded connection. The investigation determined that this fitting should be replaced on all the EDGs during the next scheduled system outage. The EDG D24 threaded pipe was replaced on 5/23/18.

On 1/21/19, the threaded connection developed a leak and the EDG D24 was declared inoperable. An investigation determined the cause was high-cycle fatigue due to the instrument line vibrating at its resonant frequency during engine operation. Contributing to this was the specific design for the piping arrangement and a U-bolt that was found broken on the 4-inch lube oil header.

The failed threaded connection was replaced, and the broken U-bolt was reinstalled on the main header. To account for the design deficiency with the instrument line piping arrangement, a clamp was installed between the main header and the instrument line to reduce vibrations until a design change could be implemented to change the instrument line configuration. Vibration readings taken following implementation of these actions found a significant reduction in the vibration of the instrument line; however, on 7/22/19, the threaded pipe connection failed again.

Analysis of the Event

The failures that occurred in 2013 were caused by a missing grommet on a support for the piping. The threaded connection and the grommet were replaced, and the connection performed without an issue for five years. After a proactive replacement was completed for EDG D24 the connection failed twice.

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Limerick Generation Station, Unit 2	05000-	19	002	00

NARRATIVE

The failure on 1/21/2019 was due to high-cyclic fatigue due to the instrument line vibrating at its resonant frequency. The threaded pipe was replaced and the EDG was returned to an operable status. The corrective actions were to install a clamp to support the one-quarter inch instrument line and to modify the instrument line design to reduce the vibration at the threaded connection.

On 3/19/19 during a walkdown of EDG D24 during the 24-hour endurance run, a broken U-bolt was identified on the four-inch main header in a location near the threaded connection. The U-bolt was repaired following the completion of the 24-hour endurance run.

On 4/25/19 a clamp was installed from the four-inch main header to the one-quarter inch instrument line. The running vibration data showed an improvement in the magnitude. It was believed that the cause of the 1/21/19 failure was due to the broken U-bolt, and with the repair of the U-bolt and the addition of the support, the cause of the failure had been addressed.

On 7/22/19 the threaded connection failed. Although the new support was installed, the threaded connection was still experiencing vibration, albeit at a reduced level. The threaded pipe was not replaced when the support was added. The threaded pipe had already experienced fatigue since it was installed several months prior to the installation of the new clamp. The modification to change the piping configuration had not yet been implemented.

Based on past operating history, threaded pipe failures have occurred during engine starts and have been known to worsen during engine runs following crack initiation. Since the failure that occurred on 7/22/19 was a sudden shear of the threaded pipe with no signs of leakage prior to failure, it is concluded that the crack initiated during the engine start on 7/22/19 and the rapid propagation of the crack occurred during the subsequent engine run-time. Therefore, D24 EDG was conservatively determined to be inoperable following the completion of the last run of the EDG on 6/18/19.

Safety Consequences

The EDG and Auxiliary Systems are safety related standby emergency power systems and consist of four diesel generator sets per Unit. The operability of the A.C. power sources and associated distribution systems during operation ensures that sufficient power will be available to supply the safety-related equipment required for (1) the safe shutdown of the facility and (2) the mitigation and control of accident conditions within the facility. The TS action requirements specified for the levels of degradation of the power sources provide restriction upon continued facility operation commensurate with the level of degradation.

The D24 EDG was conservatively determined to be inoperable from 6/18/19 through 7/25/19 therefore, the TS 3.8.1.1.a action to restore the diesel within 30 days was not met. There were minimal safety consequences associated with the condition since there were no events during this time period which required the D24 EDG. The remaining LGS Unit 2 EDGs were only inoperable but available for surveillance testing during this period.

Cause of the Event

The cause of the failure of the threaded pipe connection was due to high-cyclic fatigue. A contributing cause was the modification to change the piping configuration had not been installed prior to the failure.

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		19	- 002	- 00

NARRATIVE**Corrective Actions**

1. A temporary configuration change was implemented to remove the threaded pipe and valve, plug the lube oil piping hole, and cap off the small diameter copper instrumentation tubing. This temporary change eliminates the threaded pipe vulnerability in this location.
2. A design change is being implemented to change the configuration of the connection between the four-inch main header and the one-quarter inch instrument line.

Previous Similar Occurrences

On 3/21/13, a 1-2 drop per minute (DPM) leak was identified on the threaded connection for the EDG D24. On 4/27/13, the 1-2 DPM threaded connection leak became a 140 DPM leak, resulting in EDG D24 inoperability. The investigation performed for this failure determined that the cause to be high-cycle fatigue from engine vibration exacerbated by a missing grommet on one of the downstream supports.

On 1/21/2019, the threaded connection developed a leak and the EDG D24 was declared inoperable due to a leak on the fitting that was replaced on 5/23/18. The replacement was a proactive replacement as part of an extent-of-condition action from a failure of this threaded connection on a different EDG.

System: EK Emergency On-Site Power Supply System
Component: PSF Pipe Fitting
Manufacturer: N/A